Gyro-Aided Epipolar Filtering for Visual-Inertial Odometry on KITTI and EuRoC: An Empirical Study with Failure Analysis

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***Abstract*—Visual-inertial odometry (VIO) combines a camera with an inertial measurement unit (IMU) to estimate motion at a cost low enough for embedded platforms. We describe a compact error-state Kalman filter pipeline, built and evaluated step by step in a notebook environment on two datasets: KITTI raw (INS-grade IMU, 10 Hz camera) and a 4× downscaled copy of EuRoC (MEMS IMU, 20 Hz camera). The filter uses a gyro-predicted rotation and a two-view epipolar translation-direction measurement, plus a zero-velocity update for stops. Along the way we repair timestamp faults in the KITTI OXTS log, verify camera–IMU consistency quantitatively (median vision–gyro rotation disagreement of 0.087° per frame against 0.89° of motion), and show why uncorrected dead reckoning fails (77–516 m ATE within 70 s on EuRoC, even after static gyro-bias removal). The filter beats a never-moving baseline on 3 of 5 EuRoC sequences (ATE 0.28–0.80 m against 1.8–2.1 m) and fails on the other two. On KITTI it recovers from a wrong initial velocity on all five drives, but from an exact start it improves on IMU-only dead reckoning on only two. A zero-velocity update reduces end error on the one clear stop in our data from 2.2 to 0.3 m, with thresholds fitted to that same stop. We report every configuration tried, the hypotheses the data refuted, and the failures we could not explain.**

***Index Terms*—visual-inertial odometry, error-state Kalman filter, epipolar constraint, zero-velocity update, KITTI, EuRoC**

I. Introduction

An IMU provides high-rate, metric-scale motion and a gravity reference, but integrating it drifts without bound. A camera provides drift-limiting geometric constraints, but a single camera cannot observe scale. Fusing the two is the basis of visual-inertial odometry [1], [6], [7], and its cost and latency matter most on mobile and embedded hardware, where compute, memory and capture-path overheads are constrained.

This paper reports a small, fully instrumented study of that fusion. We built a filter-based pipeline from the data layer upward, and at each stage measured what the stage costs and what it gains against ground truth. The pipeline is a research prototype. Its camera update is a two-view relative-pose update and not a multi-view sliding-window update, and we make no claim of matching established systems such as OpenVINS [5]. The value of the study lies in the diagnostics and the complete record of results, including failures.

A. Objectives

O1 Build a reproducible VIO pipeline whose data source sits behind a time-ordered replay interface, so that recorded data can later be replaced by a live camera and IMU source.

O2 Quantify how fast inertial dead reckoning drifts on INS-grade and MEMS-grade IMUs, to establish what the camera must repair.

O3 Verify camera–IMU consistency (extrinsics, bias, frame alignment) quantitatively *before* fusing the two.

O4 Evaluate a gyro-aided epipolar update and a zero-velocity update on two datasets and two initialisation conditions.

O5 Report all outcomes, including failures, and state which hypotheses the data refuted.

B. Contributions

(i) A modular pipeline with a replay interface, undistortion, a KLT tracker with per-stage timing, and a 15-state error-state EKF with a clone of the previous camera pose. (ii) Data-quality findings: a timestamp fault in the KITTI OXTS log and its repair, and an inconsistent sequence in a cleaned EuRoC copy. (iii) Quantitative consistency checks that fix the extrinsic chain, the gyro bias and the frame alignment before any fusion. (iv) A direction-only update with three outlier-handling modes and a variance-based zero-velocity update. (v) Full reporting over 5+5 sequences, including three failures and eight refuted hypotheses.

II. Related Work

A. Filter-based VIO

The multi-state constraint Kalman filter (MSCKF) [1] keeps a sliding window of camera poses in an EKF and uses feature tracks as constraints on those poses, projecting out the landmark errors so that cost grows linearly with the number of features. Later work showed that naive EKF linearisation makes the filter inconsistent, because spurious information appears in unobservable directions (global yaw and position), and proposed observability-constrained fixes and first-estimate Jacobians [2], [3]. ROVIO [4] instead runs an EKF directly on photometric patch errors. OpenVINS [5] is a research platform built on the MSCKF with online calibration, optional SLAM landmarks and first-estimate Jacobians, and it is the closest reference point for this work in structure and in the datasets used for evaluation.

B. Optimisation-based and learned VIO

Keyframe-based fixed-lag smoothing (OKVIS [6]) and tightly coupled monocular estimation with loop closure (VINS-Mono [7]) solve nonlinear least-squares problems over a window of states. They rely on IMU preintegration on the manifold [9]. ORB-SLAM3 [8] adds visual-inertial maximum-a-posteriori estimation with multi-map support. Learned approaches such as VINet [10] regress motion from image and IMU sequences end to end. We use none of these; they bound what a filter of our size can be expected to do.

C. Error-state formulation

The error-state (indirect) Kalman filter keeps a nominal state on the manifold and estimates a small error state [11]. It avoids over-parameterised rotation states and is the formulation we use.

D. Front-end and geometry

We track corners [12] with pyramidal Lucas–Kanade flow [13], reject outliers with a forward–backward check and RANSAC [14], and use epipolar geometry [15]. The five-point relative-pose solver [16] is the standard way to get rotation and translation direction from calibrated correspondences. When rotation is known from a gyro, the epipolar constraint becomes linear in the translation direction; we use that simplification, which is standard, and do not claim novelty for it.

E. Datasets, evaluation, calibration, and stops

KITTI [17] provides outdoor driving with a high-grade GPS/INS unit; EuRoC [18] provides indoor micro-aerial-vehicle flights with a MEMS IMU and motion-capture ground truth; TUM-VI [19] is a further VIO benchmark. Trajectory evaluation conventions (alignment, absolute and relative errors) follow [20]. Temporal and spatial calibration of camera–IMU rigs is addressed by [21]. Zero-velocity updates, popularised in pedestrian navigation [22], constrain drift whenever a stationary detector fires.

F. Positioning

This work does not propose a new estimator. It contributes a worked, transparent evaluation of a small filter on two standard data sources, with the failures kept in.

III. Method

A. Data layer and timestamp repair

Both datasets are loaded as a `Sequence` object holding frame timestamps, IMU samples, ground truth and calibration. A `replay()` generator yields IMU samples and frames in time order (IMU first on ties). This is the interface a live capture source would implement.

KITTI. We use the "extract" version of the raw drives, which keeps the 100 Hz OXTS log separate from the roughly 10 Hz camera and leaves images unrectified at 1392×512. The OXTS timestamps of drive 0005 contain a fault between samples 1030 and 1063: successive intervals of $+130$, -50, and $+70$ ms, with timestamps going backwards. We fit a linear trend against the sample index, flag samples whose residual from a running median exceeds 15 ms, and replace each flagged cluster (plus one anchor on each side) by linear interpolation. After repair the intervals lie between 9.77 and 10.29 ms. The gyro step inside the repaired stretch (maximum 0.0130 rad/s) is no larger than the 99th percentile over the drive (0.0134 rad/s), so only the timestamps were corrupted.

EuRoC. We use a "cleaned" copy distributed as a Kaggle dataset. Images are downscaled $4\times$ to 188×120. The IMU is packed as ten samples per camera interval with no timestamps, which we space evenly within each interval. Ground truth is position only. The frame list of sequence V2_03 contains twelve index gaps (up to 9.6 s) and does not match the array lengths, so we exclude it and use the remaining five sequences.

B. Calibration

For KITTI we read the raw intrinsics and distortion from the calibration file ($f \approx 958$ px, $k_1 = -0.369$; the distortion displaces the image corners by 227–263 px) and undistort with a precomputed remap, keeping the original K. The camera-from-body rotation R_{cb} chains the IMU-to-velodyne, velodyne-to-camera-0, and camera-0-to-camera-2 rotations. For EuRoC we scale the published cam0 intrinsics by $1/4$ (principal point $c' = (c + 0.5)/4 - 0.5$) and undistort tracked *points*, not images.

C. Front-end

Corners are detected with a minimum-eigenvalue criterion, tracked with a pyramidal KLT tracker, validated by a forward–backward check (1 px) and a fundamental-matrix RANSAC (1 px), and kept away from the image border.

D. State and propagation

The nominal state is (R, v, p, b_g, b_a) with

$$\dot{p} = v, \quad \dot{v} = R(\tilde{a} - b_a) - g, \quad R_{k+1} = R_k \operatorname{Exp}\big((\bar{\omega} - b_g)\Delta t\big), \tag{1}$$

where $g = (0, 0, 9.81)^\top$ and $\bar{\omega}$ is the midpoint gyro rate. The error state is $\delta x = (\delta\theta, \delta v, \delta p, \delta b_g, \delta b_a)$ with $R = \hat{R}\operatorname{Exp}(\delta\theta)$. At each camera time the filter stores a stochastic clone (R_c, p_c) of the pose and its cross-covariance, giving a 21-dimensional joint error state.

E. Camera update

Direction measurement. Given tracked correspondences (unit rays x_1^i, x_2^i) and a rotation $\hat{R}$ between the two frames, the epipolar constraint is $t \cdot (\hat{R}x_1^i \times x_2^i) = 0$. The translation direction t is the right singular vector of the stacked cross products with the smallest singular value, and its sign is fixed by cheirality. On EuRoC, $\hat{R}$ is the gyro-predicted rotation (integrated with the estimated bias and mapped through R_{cb}). Pairs are used only if the median parallax is at least 0.15° and at least 70% of depths are positive.

Model. The camera centre moves by $\delta = (p - p_c) + (R - R_c)\,l$, where l is the camera position in the body frame (lever arm). The predicted unit direction in the previous camera frame is $\hat{u} = R_{cb}R_c^\top\delta/\|\delta\|$. With c the measured direction

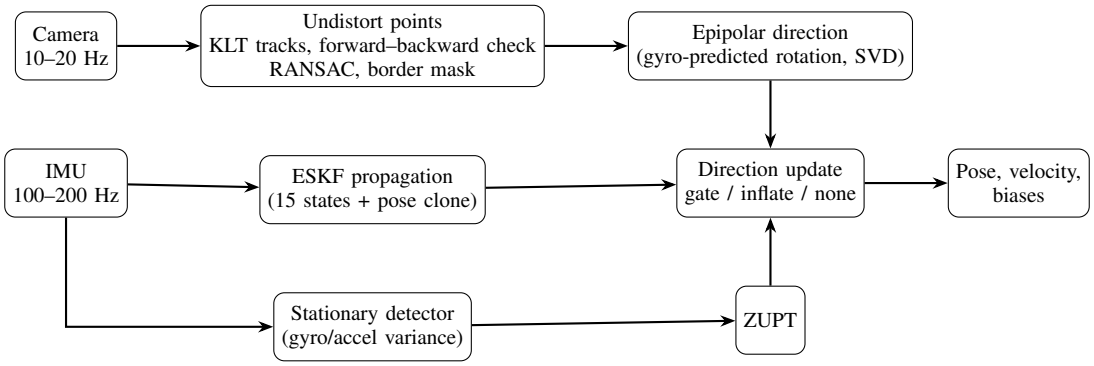


Fig. 1. Pipeline overview. Both data sources sit behind a time-ordered replay interface.

and $B \in \mathbb{R}^{3\times 2}$ an orthonormal basis of the plane normal to c, the residual is

$$r = B^\top (c - \hat{u}), \qquad \mathrm{Cov}(r) = \sigma_{\mathrm{dir}}^2 I_2. \tag{2}$$

The Jacobians follow by the chain rule through $\hat{u}$. On KITTI, we additionally use the relative rotation from an essential-matrix decomposition ($\sigma_R = 3 \times 10^{-3}$ rad) as a second measurement.

Outlier handling. The normalised innovation squared (NIS) is compared with the 99% chi-square bound (9.21 for two degrees of freedom). We compare three modes: *gate* (reject if above the bound), *inflate* (scale the noise by NIS/9.21 so the update is down-weighted and not discarded), and *none*.

F. Zero-velocity update

When the stationary detector fires, we apply the pseudo-measurement $z = -v$ with noise $\sigma_v^2 I_3$. The detector declares a stop when, over the 0.3 s of IMU samples before a frame, the root of the summed per-axis variances is below $\tau_g = 1.4 \times 10^{-3}$ rad/s (gyro) and $\tau_a = 0.05$ m/s^2 (accelerometer) for three consecutive frames.

IV. Experimental Setup

Metrics. For EuRoC we report the absolute trajectory error (ATE) after a 4-DoF alignment (yaw and translation; gravity makes roll and pitch observable) [20]. Because that alignment is generous in a small room, we always report a *stay-still* baseline: the ATE of an estimate that never moves, which is the RMS spread of the ground truth about its centroid (1.7–2.1 m here). For KITTI the start pose is known, so we report unaligned end error and RMSE.

Initialisation. KITTI starts from the ground-truth pose and velocity (attitude prior $\sigma_\theta = 0.01$ rad by default, velocity prior 0.1 m/s). To test recovery we also start with the velocity scaled by 0.6 and a velocity prior of 1.5 m/s. EuRoC starts from a gravity-aligned attitude (from the still period), zero velocity (the sequences start at rest) and the static gyro bias estimated over that period.

Protocol caveats. All five EuRoC and five KITTI sequences were examined while the filter was being developed. The default settings reported below were chosen after seeing all of them, so there is no held-out set. We report the full grids so that the reader can see the choice. All runs are deterministic: three spot-checked numbers were reproduced to the printed precision after the notebook was re-run.

V. Results

A. Consistency checks before fusion

KITTI. Over 159 frame pairs, the median rotation per frame is 0.888° from essential-matrix vision and 0.864° from the gyro. With the chained extrinsics the median disagreement is 0.087° (90th percentile 0.189°); with no extrinsics it is 1.30°, larger than the motion itself. The two possible conventions for the final camera-to-camera rotation are indistinguishable (0.087° and 0.090°). This test is sensitive mainly to yaw, since the vehicle rotates almost only about the vertical axis, so errors about the other axes would not show.

EuRoC. Essential-matrix rotation is poor at this resolution and baseline (median disagreement 0.90° against 1.46° of motion, with only 6–18% of pairs usable). We therefore test the gyro instead: rotate the previous rays by the gyro-predicted rotation and measure the epipolar residual of the correspondences (Table II). Using the gyro with its static bias removed and the extrinsic rotation lowers the residual sevenfold, from 0.141° to 0.019°. Dropping the extrinsics, using the raw gyro, or shifting the IMU window by one frame each increases the residual by a factor of 2–5. Treating the points as already undistorted is worse than undistorting them (0.051° against 0.019°), so this copy contains raw (distorted) images. The test cannot resolve shifts smaller than one frame; because we synthesise the IMU timestamps within each camera interval, there is no independent time offset to estimate.

B. Inertial dead reckoning

On KITTI drive 0005, integrating the OXTS IMU from the exact initial state gives 0.88 m ATE in 16.5 s (median 0.21 m over 5 s windows). On the EuRoC MEMS IMU the result is very different (Table III). Without bias estimation the ATE reaches 8.7–23.5 km over the full sequences. Removing the static gyro bias (about 0.08 rad/s in magnitude, mostly about the vertical axis, consistent across sequences) reduces this by 10–30×, yet 0.7–1.9 km remains. A back-of-envelope

TABLE I
DATA USED.

Dataset	Rates	Images	Sequences used
KITTI raw (extract)	camera 9.7 Hz, IMU 100 Hz	1392×512, unrectified, gray	drives 0005, 0001, 0011, 0014, 0019 (12–50 s, 74–415 m)
EuRoC (cleaned)	camera 20 Hz, IMU 200 Hz (synthesised stamps)	188×120	V1_01, V1_02, V1_03, V2_01, V2_02 (first 70 s)

TABLE II
EUROC V1_01: MEDIAN EPIPOLAR RESIDUAL (372 PAIRS WITH GYRO ROTATION $\geq 0.5°$).

Rotation used	Residual
none (identity)	$0.141°$
gyro, bias-corrected, extrinsics	$0.019°$
gyro, bias-corrected, identity R_{cb}	$0.098°$
gyro, raw (no bias correction)	$0.057°$
gyro, bias-corrected, window shifted ± 1 frame	$0.044°$ / $0.038°$
same as best, points treated as pinhole	$0.051°$

TABLE III
EUROC IMU-ONLY DEAD RECKONING, FULL SEQUENCES (ATE, M).

Sequence	zero bias	static gyro bias	5 s-window median
V1_01	23527	808	15.9
V1_02	8669	794	42.6
V1_03	16347	706	27.7
V2_01	9969	1492	50.6
V2_02	12278	1889	65.9

check supports the mechanism: a tilt error from a horizontal gyro bias b leaks gravity into horizontal acceleration, giving position error of order $g\,b\,t^3/6$; with the measured horizontal bias of 0.023 rad/s and $t = 143$ s this is about 100 km, the same order of magnitude as the unaligned end error (75 km) of the zero-bias run on V1_01. We measured noise densities of 1.5×10^{-3} rad/s/$\sqrt{\text{Hz}}$ (gyro) and 3.8×10^{-2} m/s^2/$\sqrt{\text{Hz}}$ (accelerometer) from the still periods, about 9 and 19 times the datasheet-style defaults we started from. A narrow spectral peak at 72–88 Hz holds 23–54% of the accelerometer power in four sequences, so part of this is probably platform vibration and not sensor noise; we cannot separate the two.

C. Front-end behaviour and latency

On KITTI the tracker keeps a median of 273 inlier tracks per frame (minimum 182), but tracks are short: mean length 3.4 frames, median 1. Of the features lost, 58% fail the forward–backward check or the KLT status flag, 30% fail RANSAC, and only 12% leave the border. The loss rate rises with distance from the principal point (9.5% within 150 px to 57% beyond 600 px) and toward the bottom of the image (64% in the lowest 128 rows), and 69% of newly detected features die at their first tracking step. A larger search (five pyramid levels, 31 px window) does not help (280 tracks, mean length 3.5) and raises the cost from 22 to 35 ms; stricter corner quality changes mean length only from 3.4 to 3.9. The cause is unresolved; one untested candidate is appearance change on the road surface under forward motion. On EuRoC at the lower resolution, tracking is healthy (median 104 tracks, mean length 29 frames), and 56% of losses are features leaving the image.

TABLE IV
PER-STAGE LATENCY, MS (MEDIAN / 99TH PERCENTILE).

Pipeline	Stage	p50 / p99
KITTI 1392×512	undistort remap	1.28 / 1.95
	KLT	12.85 / 20.18
	RANSAC	0.64 / 1.36
	detection	11.69 / 14.31
	tracker total	25.46 / 34.21
EuRoC 188×120	tracker total	3.13 / 3.81
	epipolar direction	0.58 / 0.97
ESKF propagate	per IMU sample, μs	158.7 / 287.9

Table IV gives per-stage latency on a shared 4-vCPU Intel Xeon at 2.0 GHz (single run; repeat runs differed by about 15%). The KITTI figure excludes the essential-matrix solve and the filter update. Propagation costs 159 μs per IMU sample in NumPy, about 3% of one core at 200 Hz.

D. EuRoC

Table V summarises the first 70 s of each sequence with the default setting (inflate, $\sigma_{\text{dir}} = 0.3$). The filter beats the stay-still baseline on V1_01, V2_01 and V2_02 (ATE 0.28, 0.80 and 0.74 m against 1.85, 2.06 and 2.09 m: 2.6× to 6.7× lower) and is vastly better than IMU-only on every sequence. It does not beat standing still on V1_03 (2.42 against 1.74 m) and it diverges on V1_02 (1480 m).

The full grid (Table VI) shows that results are not smooth in the noise setting. With a hard gate, V1_02 and V1_03 diverge for most directions of sigma, and the runs that diverge reject 70–95% of their updates. Down-weighting instead of discarding (*inflate*) or removing the gate (*none*) fixes V1_03's divergence (from 535–666 m to 2.2–2.6 m) and removes a sensitivity on V2_01 (6.85 m at $\sigma = 0.15$ with gating, at most 1.44 m otherwise), but **it does not rescue V1_02**. At $\sigma = 0.5$ no update is rejected in any mode, and V1_02 still ends at 1133 m in all three. The gate is therefore not the failure mechanism for V1_02.

We looked at V1_02 in detail. In three runs the speed estimate leaves the ground-truth speed at the same moment (6.8–7.2 s), about 3 s after the still period ends, and in the diverging runs it then gains roughly 1–3 m/s^2 on average. The estimated accelerometer bias stays at 0.2–0.4 m/s^2, too small to explain that growth. A tilt error of 0.1–0.3 rad could, but the cleaned copy has no orientation ground truth, so we cannot test it. Compared with the other sequences, V1_02 has the highest median speed (0.98 m/s against 0.31–0.78) and V1_02, V1_03 and V2_02 all rotate at about 0.55 rad/s, yet V2_02 works,

TABLE V
EUROC, FIRST 70 S, ATE IN METRES (DEFAULT: INFLATE, $\sigma_{\mathrm{dir}} = 0.3$).

Seq.	stay still	IMU only	VIO (gate)	VIO (inflate)
V1_01	1.85	76.7	0.27	0.28
V1_02	1.84	516.0	1359.4	1480.5
V1_03	1.74	255.1	2.41	2.42
V2_01	2.06	475.0	0.44	0.80
V2_02	2.09	475.7	0.75	0.74

so no single motion statistic separates the failures from the successes.

Two further EuRoC findings are negative. Widening the bias priors does nothing while updates are being rejected; what mattered was the direction noise (at $\sigma_{\mathrm{dir}} = 0.05$ almost every update is gated, whereas 0.15 accepts 96%), because the predicted direction drifts quickly enough that a tight gate discards good measurements. And modelling the 7 cm camera–IMU lever arm had no measurable effect on V1_01 (0.42 m with it, 0.39 m without).

E. KITTI

Tables VII and VIII give end error and RMSE for six configurations: IMU-only and five variants of the direction/rotation update (A–E, differing in the initial attitude prior and σ_{dir}). With the default configuration A and a corrupted initial velocity, vision lowers the end error on all five drives (for example 295 to 31 m on drive 0019 and 22 to 2.1 m on drive 0005). With an exact start the picture is mixed: against IMU-only, configuration A is better by end error on two drives (0001, 0019) and worse on three (0005, 0011, 0014), and the largest degradation is on drive 0014 (12.7 to 68.4 m).

For drive 0014 we found the proximate mechanism. The speed estimate creeps upward (20.4 against 14.9 m/s at the end), and the creep rate (mean $+0.146$ m/s^2) matches the along-track acceleration error implied by a persistent pitch error of 0.5–0.9° ($g \sin\theta \approx +0.137$ m/s^2). The creep scales with the trust placed in the direction measurement: end speed is 19.6, 18.0, 16.3 and 15.1 m/s for $\sigma_{\mathrm{dir}} = 0.05, 0.1, 0.2, 0.5$ (truth 15.06). Tightening the attitude prior helps (end error 68.4 to 38.8 m at $\sigma_\theta = 0.001$) but does not remove it. Perturbing the extrinsic rotation by $\pm 0.5°$ and $\pm 1°$ about each axis does not consistently help (the best perturbation differs on every drive and leaves 0014 at 66–72 m), so a simple extrinsic error is ruled out. No configuration is best everywhere: larger σ_{dir} (D, E) gives the best exact-start results on 0011 and 0014 and the worst recovery from the wrong start (0011: 74 and 104 m against 18 m for configuration C).

F. Zero-velocity update

Only KITTI drive 0011 contains a clear stop (the car is stationary for the last 12 s); drives 0005, 0001 and 0014 have none, and on 0019 the vehicle only slows to 0.09 m/s. On 0011, gyro and accelerometer spreads separate stationary from moving windows with a thin margin (stationary 95th percentile 0.0014 rad/s against a moving 5th percentile of 0.0017 rad/s; 0.052 against 0.047 m/s^2), and the combined detector admits 2.6% of moving windows. On EuRoC no usable detector exists: the accelerometer spread of stationary windows (0.34–0.63 m/s^2) overlaps the moving windows, which pass 57–93% of the time; the gyro rejects moving windows well on two sequences (0.0% and 0.5% false accepts) and poorly on three (31–41%).

On KITTI the detector flags 39 of the 46 truly stopped frames on 0011 and fires on none of the frames of the three other drives with no stop. On 0019 it fires on 35 frames during a slow roll, 16 of them with ground-truth speed above 0.2 m/s (maximum 0.38 m/s). No threshold pair we tried removes those without cutting the 0011 hits (the tightest pair left 12 false frames and one hit), which follows from the physics: an accelerometer cannot distinguish a steady creep from rest. We therefore set σ_v by a rule fixed before looking at the results: adopt 0.2 m/s unless it more than doubles the end error on 0011 from the wrong start relative to 0.05 m/s. It did not (10.8 against 14.7 m for IMU-only; 13.2 against 12.8 m for VIO).

Table IX shows the outcome. On 0011 the update cuts end error from 2.2 to 0.3 m (IMU-only) and from 3.0 to 0.2 m (VIO) from the exact start, and from 119 to 11 m and 84 to 13 m from the wrong start. The size of the second improvement exceeded our expectation (we predicted about 60 m). One explanation consistent with the numbers, which we did not test directly, is that for a pure initial-velocity error the filter's covariance ties the position error to the velocity error, so correcting the velocity at the stop also corrects the position; real drift from biases and noise lacks that structure, so the wrong-start case is probably the most favourable one for this update. On 0019, ZUPT helps IMU-only (17.8 to 2.4 m) but not VIO, where RMSE rises from 5.5 to 7.2 m and end error from 11.8 to 15.8 m; a looser σ_v did not close the gap.

VI. DISCUSSION

A. Hypotheses the data refuted

Several of our own explanations failed, and we list them because the corrections shaped the pipeline.

1) A larger KLT search would lengthen KITTI tracks. It did not (mean length 3.4 to 3.5).
2) Essential-matrix rotation is adequate at 188×120. It is not (median error $0.90°$ against $1.46°$ of motion); the gyro supplies rotation instead.
3) Wider bias priors fix the EuRoC gating failures. They do not; the direction noise did.
4) An extrinsic rotation error of up to $1°$ causes the speed creep on KITTI 0014. No consistent improvement for any axis.
5) The camera–IMU lever arm matters on EuRoC. No measurable effect on V1_01.
6) The hard gate causes the V1_02 divergence. With no rejections at all it still diverges.
7) After a zero-velocity update from a wrong start, about 60 m of error would remain on 0011. About 11–13 m remained.

TABLE VI
EuRoC: every setting tried, ATE in metres, by outlier mode and direction noise σ_{dir} (rad).

Seq.	Mode	0.1	0.15	0.3	0.5
V1_01	gate	0.93	0.42	0.27	0.20
	inflate	0.42	0.41	0.28	0.20
	none	0.36	0.33	0.28	0.20
V1_02	gate	986.74	2.21	1359.39	1133.48
	inflate	1526.93	1599.58	1480.49	1133.48
	none	5.92	2037.31	1489.11	1133.48
V1_03	gate	666.00	534.74	2.41	23.13
	inflate	2.15	2.33	2.42	23.13
	none	2.64	2.42	2.42	23.13
V2_01	gate	0.46	6.85	0.44	0.39
	inflate	1.04	1.07	0.80	0.39
	none	1.43	1.44	0.81	0.39
V2_02	gate	0.60	0.66	0.75	0.71
	inflate	0.76	0.81	0.74	0.71
	none	0.81	0.85	0.73	0.71

TABLE VII
KITTI, exact initial state: end error / RMSE (m), unaligned. σ_θ = initial attitude prior (rad).

Config.	$(\sigma_\theta, \sigma_{\mathrm{dir}})$	0005	0001	0011	0014	0019
IMU only	–	2.9/1.8	0.5/0.3	2.2/1.3	12.7/7.4	17.8/8.1
A (default)	(0.01, 0.05)	5.1/2.3	0.4/0.3	3.0/1.0	68.4/29.0	11.8/5.5
B	(0.003, 0.05)	5.3/2.2	0.5/0.4	2.4/0.9	39.7/15.7	7.4/4.6
C	(0.001, 0.05)	5.5/2.0	0.5/0.4	1.9/0.9	38.8/15.1	5.3/4.3
D	(0.001, 0.2)	5.0/1.9	0.4/0.3	1.0/0.6	6.3/3.0	4.7/3.7
E	(0.01, 0.5)	4.2/1.7	0.4/0.3	0.9/0.5	4.3/3.3	5.7/3.7

TABLE VIII
KITTI, initial velocity $\times 0.6$ (prior 1.5 m/s): end error / RMSE (m), unaligned.

Config.	$(\sigma_\theta, \sigma_{\mathrm{dir}})$	0005	0001	0011	0014	0019
IMU only	–	21.9/12.7	62.2/35.9	119.2/68.4	120.0/69.2	294.6/168.7
A (default)	(0.01, 0.05)	2.1/1.5	29.0/23.1	84.1/46.6	76.2/32.2	30.6/33.2
B	(0.003, 0.05)	3.7/1.6	21.1/16.6	24.7/18.0	48.8/20.2	28.0/30.3
C	(0.001, 0.05)	5.0/1.8	16.9/14.2	17.6/13.9	46.7/19.1	24.7/26.5
D	(0.001, 0.2)	4.7/1.7	42.7/28.6	74.3/43.0	19.2/8.1	69.0/58.4
E	(0.01, 0.5)	2.5/2.1	58.2/34.1	103.5/59.7	18.3/9.2	106.9/88.3

TABLE IX
ZUPT ($\sigma_v = 0.2$ m/s): end error / RMSE (m).

Drive	Start	Config.	no ZUPT	ZUPT
0011	exact	IMU only	2.2/1.3	0.3/0.8
	exact	VIO	3.0/1.0	0.2/0.8
	$\times 0.6$	IMU only	119.2/68.4	10.8/41.5
	$\times 0.6$	VIO	84.1/46.6	13.2/30.3
0019	exact	IMU only	17.8/8.1	2.4/6.1
	exact	VIO	11.8/5.5	15.8/7.2
	$\times 0.6$	IMU only	294.6/168.7	74.2/144.2
	$\times 0.6$	VIO	30.6/33.2	39.6/34.4

8) A looser σ_v repairs the VIO penalty on 0019. It did not.

B. Threats to validity

The development and the reported sequences coincide, so every setting is in-sample; the full grids are given for that reason. KITTI runs are 12–50 s long and start from ground truth. The EuRoC copy is downscaled, has synthesised IMU timestamps and no attitude ground truth. KITTI ground truth comes from the same GPS/INS unit that supplies the IMU data. The zero-velocity thresholds were taken from the same single stop used to evaluate them. Noise parameters were measured on a single sequence's still period and probably include vibration. Latency was measured once on a shared CPU. Finally, the aligned ATE flatters any estimate in a small room, which is why every EuRoC result is paired with the stay-still baseline.

VII. Conclusion and Future Work

We built and evaluated a small filter-based visual-inertial pipeline on KITTI and EuRoC and kept the evidence for what works and what does not. Camera–IMU consistency can be verified to a small fraction of a degree before fusion. Dead

reckoning on a MEMS IMU diverges to hundreds of metres within 70 s even after static gyro-bias removal. A gyro-aided epipolar direction update beats a never-moving baseline on three of five EuRoC sequences and recovers from a wrong initial velocity on all five KITTI drives, but fails on two EuRoC sequences for reasons we have not identified, and does not improve on IMU-only from an exact KITTI start.

Future work follows the gaps. A multi-view (MSCKF-style) update using whole feature tracks should replace the two-view update, and online estimation of the camera–IMU rotation and time offset would remove the assumptions behind our consistency checks. Full-resolution EuRoC with attitude ground truth would allow testing the tilt-error explanation for the V1_02 failure. The propagation and tracking steps are natural candidates for a C++ implementation with unit tests, using the per-stage numbers here as the baseline, and the replay interface lets a live capture source replace the recorded data. Retrieval- and vision-language-model layers on keyframes, part of the original plan for this project, are not implemented.

REFERENCES

[1] A. I. Mourikis and S. I. Roumeliotis, "A multi-state constraint Kalman filter for vision-aided inertial navigation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2007.

[2] M. Li and A. I. Mourikis, "High-precision, consistent EKF-based visual-inertial odometry," *Int. J. Robotics Research*, 2013.

[3] J. A. Hesch, D. G. Kottas, S. L. Bowman, and S. I. Roumeliotis, "Camera-IMU-based localization: Observability analysis and consistency improvement," *Int. J. Robotics Research*, 2014.

[4] M. Bloesch, S. Omari, M. Hutter, and R. Siegwart, "Robust visual inertial odometry using a direct EKF-based approach," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2015.

[5] P. Geneva, K. Eckenhoff, W. Lee, Y. Yang, and G. Huang, "OpenVINS: A research platform for visual-inertial estimation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020.

[6] S. Leutenegger, S. Lynen, M. Bosse, R. Siegwart, and P. Furgale, "Keyframe-based visual-inertial odometry using nonlinear optimization," *Int. J. Robotics Research*, 2015.

[7] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Trans. Robotics*, 2018.

[8] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial and multi-map SLAM," *IEEE Trans. Robotics*, 2021.

[9] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, "On-manifold preintegration for real-time visual-inertial odometry," *IEEE Trans. Robotics*, 2017.

[10] R. Clark, S. Wang, H. Wen, A. Markham, and N. Trigoni, "VINet: Visual-inertial odometry as a sequence-to-sequence learning problem," in *Proc. AAAI Conf. Artificial Intelligence*, 2017.

[11] J. Solà, "Quaternion kinematics for the error-state Kalman filter," arXiv:1711.02508, 2017.

[12] J. Shi and C. Tomasi, "Good features to track," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 1994.

[13] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 1981.

[14] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, 1981.

[15] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.

[16] D. Nistér, "An efficient solution to the five-point relative pose problem," *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2004.

[17] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," *Int. J. Robotics Research*, 2013.

[18] M. Burri *et al.*, "The EuRoC micro aerial vehicle datasets," *Int. J. Robotics Research*, 2016.

[19] D. Schubert *et al.*, "The TUM VI benchmark for evaluating visual-inertial odometry," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2018.

[20] Z. Zhang and D. Scaramuzza, "A tutorial on quantitative trajectory evaluation for visual(-inertial) odometry," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2018.

[21] P. Furgale, J. Rehder, and R. Siegwart, "Unified temporal and spatial calibration for multi-sensor systems," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2013.

[22] E. Foxlin, "Pedestrian tracking with shoe-mounted inertial sensors," *IEEE Computer Graphics and Applications*, 2005.